Angle moments and variance

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2. Verified sharp purity interval with a fully specified detector

Let $A = Z \otimes I_2$, $B = X \otimes I_2$. Use normalized output-first Choi vectors

$$p = |I_4\rangle, \qquad u = |iA\rangle, \qquad v = |iB\rangle, \qquad |M\rangle = (M \otimes I)|\Phi_4\rangle.$$

These vectors are orthonormal. Define

$$\begin{aligned} k &= \frac{(sh - \sqrt{shq_0})^2}{sh(1-h)} = 0.115482549888153\ldots, \\ d &= s + k = 0.524857549888153\ldots, \\ r &= 1 - d = 0.475142450111847\ldots. \end{aligned}$$

$w = \sqrt{s}\,p - \sqrt{k}\,u$, $\eta = w/\sqrt{d}$, and the **fixed** detector

$$E = |e\rangle\langle e|, \qquad e = \sqrt{h}\,p + \sqrt{1-h}\,u.$$

Let $F = |v\rangle\langle v|$ be the added, orthogonal outcome, and let $D = I - P - |u\rangle\langle u| - F$, a rank-13 projector. The model's complete lower-endpoint phase curve, together with the known s, h, fixes the compression of ρ onto $\mathrm{span}\{p, u\}$ to $|w\rangle\langle w|$. This **requires the complete curve**, not merely q_0. Positivity then forces zero support on the orthogonal direction inside that span.

For $f = \mathrm{Tr}(F\rho)$, $0 \le f \le r$, the sharp purity lower envelope is

$$\boxed{J(f) = d^2 + f^2 + \frac{(r-f)^2}{13}}, \qquad \boxed{J(f) \le \mathrm{Tr}(\rho^2) \le 1}.$$

The minimum is attained by

$$\rho_\star(f) = |w\rangle\langle w| + fF + \frac{r-f}{13}D.$$

It is a mixture of normalized unitary Choi vectors, hence is CPTP and unital. A pure unitary Choi state also attains the same phase curve, f, and the specified product-input $Z = +1$ probability

$$\boxed{p_0(f) = d + \frac{6(r-f)}{13}}.$$

The source's construction uses a first-qubit Hermitian unitary and a second-qubit X, with anticommuting Pauli terms. Its unitarity, input and output Choi marginals, and probabilities can be checked by direct 16×16 multiplication. Mixing the pure state with $\rho_\star(f)$ continuously fills $[J(f), 1]$, while preserving the complete phase curve, f, and $p_0(f)$.

f	$J(f)$, minimum purity	$p_0(f)$, output first-qubit $Z = +1$
0	0.292841628282	0.744154065324

f	$J(f)$, minimum purity	$p_0(f)$, output first-qubit $Z=+1$
$r/14$	0.291601186810	0.728490028508
$r/2$	0.336257079801	0.634505807606
r	0.501235795573	0.524857549888

The two named endpoint Choi states have half-trace distance 0.766538917119 and optimal **state** discrimination success $(1+T)/2 = 0.883269458559$ at equal priors. That is discrimination of two specified normalized Choi states with a fixed input; unrestricted channel discrimination is a distinct optimization, as explained by Watrous.

3. Additional result: an exact purity-excess identity

Proposition (purity projection identity). Under the exact phase-curve and F-population assumptions of §2, every compatible density operator, with or without the Choi marginal constraint, obeys

$$\boxed{\mathrm{Tr}(\rho^2) = J(f) + \|\rho - \rho_\star(f)\|_2^2.} \tag{1}$$

Proof. Let $P_\eta = |\eta\rangle\langle\eta|$. The measured full phase response and positivity give

$$\mathrm{Tr}(P_\eta\rho) = d, \quad \mathrm{Tr}(F\rho) = f, \quad \mathrm{Tr}(D\rho) = r - f.$$

As $\rho_\star = dP_\eta + fF + [(r-f)/13]D$, the three fixed populations imply

$$\mathrm{Tr}(\rho\rho_\star) = d^2 + f^2 + \frac{(r-f)^2}{13} = \mathrm{Tr}(\rho_\star^2) = J(f).$$

Expand $\mathrm{Tr}[(\rho-\rho_\star)^2]$ and rearrange. No commutativity between ρ and $\rho_\star$ is assumed. Equality $\mathrm{Tr}\,\rho^2 = J(f)$ holds exactly at $\rho = \rho_\star$. □

This identity gives a **model-independent-within-the-specified-feasible-class** quantitative interpretation of additional measurements: deviations from the minimum-purity model are controlled by the Hilbert–Schmidt norm.

Corollary (an additional product-input probability). Let p be the first-qubit $Z=+1$ probability on product input $|00\rangle$, computed from the **same** trace-preserving channel. Then

$$\boxed{\mathrm{Tr}(\rho^2) \geq J(f) + \frac{13}{276}[p - p_0(f)]^2.} \tag{2}$$

Proof. In the output-first normalized Choi convention the probability observable is

$$M = 4(|0\rangle\langle 0| \otimes I_2)_{\mathrm{out}} \otimes |00\rangle\langle 00|_{\mathrm{ref}}.$$

Let $\theta = (\sqrt{k}\,p + \sqrt{s}\,u)/\sqrt{d}$. The exact phase curve forces $\rho\theta = 0$ and $\rho_\star\theta = 0$. Set $B_\theta = I - |\theta\rangle\langle\theta|$. The Hilbert–Schmidt projection of $B_\theta M B_\theta$ orthogonal to the three fixed-population operators P_η, F, D is

$$M_\perp = B_\theta M B_\theta - P_\eta - \frac{6}{13}D.$$

Direct finite-rank identities give $\mathrm{Tr}\,M^2 = 32$, $\langle\theta, M\theta\rangle = 1$, $\langle\theta, M^2\theta\rangle = 4$, $\langle\eta, M\eta\rangle = 1$, $\langle v, Mv\rangle = 0$, and $\mathrm{Tr}(DM) = 6$. Consequently

$$\|M_\perp\|_2^2 = (32 - 8 + 1) - 1 - \frac{36}{13} = \frac{276}{13}.$$

Since $\rho - \rho_\star$ is orthogonal to the three fixed-population operators and supported in B_θ, Cauchy–Schwarz yields

$$|p - p_0(f)| = |\operatorname{Tr}[M_\perp(\rho - \rho_\star)]| \le \sqrt{276/13}\,\|\rho - \rho_\star\|_2.$$

Insert (1). □

The coefficient in (2) is a **certified, generally non-sharp** lower bound. Trace-preservation and positivity can strengthen it further. When $p = p_0(f)$, the bound correctly reduces to $J(f)$; explicit pure and mixed channels show that equality of these rates cannot identify purity.

For finite-data intervals $f \in F_0$, $p \in P_0$, with the other input quantities known exactly, a valid population purity bound is obtained by minimizing the right side of (2) over $F_0 \cap [0, r]$ and $P_0 \cap [0, 1]$. The input quantities themselves must be jointly calibrated and their uncertainties propagated before applying this to experiment.

4. Verified angle-dependent contrast and a variance refinement

The phase-calibration source studies the **general joint-density-operator class**, without imposing a Choi marginal. Its conclusions are sharp in that class. A stronger CPTP or instrument-specific conclusion needs a separate proof.

For a coherent error angle ε about the intended reflection, put $x = \cos\varepsilon$, $a = sh$, $t = a - q_0$, $d_1 = s(1 - h)$, and define

$$b_0 = (1 - s)(1 - h), \quad u_0 = a + q_0, \quad R_0 = 2\sqrt{aq_0},$$
$$z_0 = \sqrt{4ab_0 - (t + b_0)^2} = 0.0852578917486220\ldots.$$

The source proves that if the state may depend on ε but the **same fixed effect** and exactly the same s, h, q_0 apply in every angle stratum, the sharp minimum contrast at a specified cosine is

$$m(x) = \begin{cases} (t + b_0)(1 + x) - z_0\sqrt{1 - x^2}, & -1 \le x \le x_1, \\ (t + u_0)(1 + x) - R_0\sqrt{2(1 + x)}, & x_1 \le x \le 1, \end{cases} \tag{3}$$

where $x_1 = 2[(u_0 - b_0)/R_0]^2 - 1 = 0.9753069372438004\ldots$. Its branches agree with matching first derivatives. In contrast, if a *single conditional response* applies across all realized angles, the strict-increase criterion is $1 + C > \kappa|S|$, with $\kappa = 1.775489004780283\ldots$, $C = \mathbb{E}\cos\varepsilon$ and $S = \mathbb{E}\sin\varepsilon$.

For states depending on angle, the worst-case average with an **entire specified angle law** is $\mathbb{E}m(\cos\varepsilon)$, whereas with only its first sine/cosine moments the worst case is $m(C)$, by convexity and an attaining two-point angular distribution. The sharp strict-rise condition in this broader class is

$$C > \frac{\kappa^2 - 1}{\kappa^2 + 1} = 0.5183463333912298\ldots.$$

For equal +60° and −60° offsets, the source's two population calculations reproduce:

shared response: $+0.064901942155\ldots$, angle-dependent states: $-0.001806421221\ldots$.

Further proposition (variance-strengthened guarantee). If the entire angle law is relevant to the same group-specifically calibrated instrument, put $X = \cos\varepsilon$, $C = \mathbb{E}X$ and $V = \operatorname{Var}(X)$. Then

$$\boxed{\mathbb{E}[\Delta(\varepsilon)] \geq \mathbb{E}[m(X)] \geq m(C) + \frac{z_0}{2}V.} \tag{4}$$

The constant z_0 is the **largest valid global strong-convexity constant** for the specific function m in (3).

Proof. In the first branch,

$$m''(x) = \frac{z_0}{(1-x^2)^{3/2}} \geq z_0,$$

and in the second branch,

$$m''(x) = \frac{R_0}{2\sqrt{2}(1+x)^{3/2}} \geq R_0/8 = 0.09591181732043\ldots > z_0.$$

The first derivatives match at x_1. Hence $m(x) - (z_0/2)x^2$ is convex on $[-1, 1]$. Jensen's inequality applied to that function proves the second inequality in (4); the first holds separately in every calibrated angle stratum. Because $m''(0) = z_0$, no larger uniform curvature constant holds on the full domain. □

Equation (4) uses a **measured or otherwise independently established** distribution; it does not invent such a distribution. Its variance correction can improve a lower contrast guarantee when full angle information is available. The condition $m(C) + (z_0/2)V > 0$ is sufficient, not generally necessary, for strict positivity. This refinement does not apply if only pooled s, h, q_0 are known or if the angle distribution transfers incorrectly to the gate path.